

Comparing Three Numerical Methods for Planar Two-Link Arm Motion

Course. CSE402: Numerical Analysis, Simulation and Modeling Sessional

Section. C-2

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Problem Statement & Scope

Problem. Compute $q(t), \tau(t)$ on a fixed horizon $[0, T]$ taking a planar two-link arm from pose A to pose B . With $q = (\theta_1, \theta_2)^\top$ and $\tau = (\tau_1, \tau_2)^\top$,

$$\begin{aligned} M(q) \ddot{q} + C(q, \dot{q}) \dot{q} + g(q) &= \tau, & t \in [0, T], \\ q(0) &= q_A, \quad q(T) = q_B, \\ q_{\min} &\leq q(t) \leq q_{\max}. \end{aligned}$$

Approach (course directions).

- ▶ **Cross-domain application:** course numerical tools (time integration, collocation, solving nonlinear systems) applied to robot-arm motion.
- ▶ **Exploratory / comparative study:** the same model and task, three established methods, compared on accuracy and cost. Not a new algorithm.

Scope

Simulation only (Python). Two-link planar arm. We do not reproduce the base paper's high-DoF pipeline or use hardware.

Base Paper Overview

Cardona-Ortiz, D. and Arechavaleta, G. (2025).

Trajectory optimization for highly articulated robots based on sparsity-free local direct collocation.

International Journal of Applied Mathematics and Computer Science, 35(4), 577–589.

<https://doi.org/10.61822/amcs-2025-0041>

Domain. Numerical optimal control for robot motion (many-joint arms and similar mechanisms).

Summary. The paper plans robot motions by direct collocation: trapezoidal and Hermite–Simpson rules turn continuous physics into a finite set of equations, which are then solved numerically. Their extra software (sparse Jacobian assembly, Pinocchio, high-DoF platforms) is not our implementation target.

Method recipes we follow in code are those in Kelly, *An Introduction to Trajectory Optimization*, SIAM Review, 2017.

Expected Methodology

Model. Planar two-link arm; states are joint angles and velocities; inputs are joint torques.

Methods (identical start, goal, and time budget).

1. Shooting with fourth-order Runge–Kutta integration
2. Trapezoidal direct collocation
3. Hermite–Simpson direct collocation

Evaluation. Tip error at the goal, residual of the dynamics, and runtime, versus number of time nodes.

Deliverables. Implementation; comparison table and plots; synchronized matplotlib visualization of the three motions.